

Jeremy Lu

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Experience

Data Analyst, NYU Langone Health Rapid RCT Lab – New York, NY May 2023 – Present

- Present quality improvement trial ideas and results to leadership teams ranging from C-Suite, kidney transplant, antibiotic stewardship, and other areas of hospital.
- Assessed relationship between high heat days and 200,000 emergency department visits over past four years using case crossover design, in order to build new patient safety messaging system.
- Evaluated 4200 encounters over 15 months from an antibiotic stewardship program using difference in differences methods and found that it reduces 30 day readmission rate by 20%.
- Built gravity model using six years of historical data totaling over a million emergency department visits to evaluate catchment area and inform patient capacity and recruitment decision making in NYC neighborhoods.

AI Research Intern, Aizip, Inc. – Cupertino, CA June 2022 – Aug 2022

- Automated process for deploying imagenet-based PyTorch Convolutional and Recurrent Neural Networks on hardware.
- Managed 3 undergraduate interns to collect, label, and organize visual and audio data for multiple TinyML neural networks in IoT.

Research Assistant, UW Department of Atmospheric Sciences – Seattle, WA Sept 2019 – Jun 2021

- Performed exploratory analysis on over 140,000 particles from Southern Ocean Clouds, Radiation, Aerosol Transport Experimental Study (SOCRATES) probe data using matplotlib, numpy, and xarray.
- Used Scikit-Learn to build and optimize a binary random forest classifier, distinguishing over 5 million liquid and ice cloud particles with 95% accuracy, a 20% increase from previous non-ML methods.

Technologies

Languages: Python (pandas, matplotlib, plotly, numpy, scikit-learn, seaborn, pytorch, onnx, flask), R (base, tidyverse, ggplot, Shiny, glm, stan, bartcause), Java, HTML, CSS, SQL, SQLite

Software: Jupyter Notebook, Amazon Web Services, Git, Microsoft Office, LaTeX, Adobe Acrobat, Tableau

Projects

NBA Lineup Finder ([Webpage](#)) March 2025

- Displays advanced lineup data from NBA.com API with custom filters for user inputted team selection, player inclusion, exclusion and minute restrictions.
- Web implementation of NBA Lineup Scraper using AWS Lambda and Amazon S3 for a serverless design, with AWS API Gateway integration to securely handle requests and responses between the web app and backend.
- Utilized ChatGPT to streamline frontend design in HTML, CSS, and JavaScript.

Education

New York University, M.S. in Applied Statistics for Social Science Research Sept 2021 – May 2023

University of Washington - Seattle, B.S. in Applied and Computational Mathematical Sciences (Data Science Concentration) Sept 2018 – Jun 2021

Publications

The University of Washington Ice-Liquid Discriminator (UWILD) improves single-particle phase classifications of hydrometeors within Southern Ocean clouds using machine learning Nov 2021

Rachel Atlas, Johannes Mohrmann, Joseph Finlon, *Jeremy Lu*, Ian Hsiao, Robert Wood, Minghui Dao
[10.5194/amt-14-7079-2021](https://doi.org/10.5194/amt-14-7079-2021)